ROTATION ANALYTICS INC

Fatigue Risk Analysis: Mathematical Methodology and Validation

A Technical White Paper on the Biomathematical Fatigue Model Used by Rotation Analytics

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1. Executive Summary

Rotation Analytics applies a biomathematical fatigue model to rotation schedules to identify where schedules create physiological fatigue risk — even when fully compliant with collective agreements. The model is based on the Three-Process Model of alertness regulation, originally published by Torbjörn Åkerstedt and Simon Folkard in 1987 and refined through 2008.

The model produces a Fatigue Score on a 0–100 scale for every day of the rotation for every rotation line. Scores are classified into four risk levels — Low, Moderate, High, and Critical — enabling stakeholders to identify schedule segments that create meaningful fatigue exposure and make informed decisions about rotation design.

This white paper documents the complete scientific foundation, mathematical methodology, engine architecture, calibration decisions, model parameters, assumptions, and limitations of the fatigue analysis delivered by Rotation Analytics.

2. Scientific Foundation

The Three-Process Model is the most widely cited and validated biomathematical model for shift work fatigue in the occupational health literature. Originally published by Åkerstedt and Folkard in 1987, the model has been revised and validated through 2008 across multiple occupational settings, including healthcare, transportation, and industrial operations.

The model was selected for this implementation because:

- It is published openly in peer-reviewed scientific literature — its equations and parameters are transparent and auditable
- It was developed specifically to model shift work environments
- Its parameters have been validated against real worker performance data, including healthcare and industrial settings
- It can be implemented deterministically without proprietary software
- It produces outputs that are explainable to clients, regulators, and arbitrators

The published model describes three independent physiological processes: homeostatic sleep pressure (Process S), circadian rhythm (Process C), and sleep inertia (Process W). The Rotation Analytics implementation activates Process S and Process C as the primary fatigue drivers, with a calibrated sleep debt modifier. Process W (sleep inertia) is documented below but is zeroed in the current implementation, as explained in the Calibration Notes.

3. The Three Processes

Human alertness is the product of interacting physiological systems operating simultaneously. The fatigue engine calculates each independently at every time step across the rotation.

3.1 Process S — Homeostatic Sleep Pressure

The longer a person is awake, the more their body builds pressure to sleep. This is a measurable neurochemical process — adenosine accumulates in the brain during wakefulness and is cleared during sleep. Process S models this mechanism as an exponential function that rises during wakefulness and falls during sleep.

During wakefulness, sleep pressure increases toward an upper asymptote of 1.0:

```
S(t) = S(t-1) + (1 - S(t-1)) × (Δt / τ_w)
```

During sleep, sleep pressure decays toward 0:

```
S(t) = S(t-1) - S(t-1) × (Δt / τ_s)
```

Parameters:

```
τ_w = 18 hours (wake time constant — rate of pressure buildup)
τ_s = 4 hours (sleep time constant — rate of pressure dissipation)
```

The asymmetry between these time constants is critical: pressure accumulates slowly across a shift (18-hour time constant) and clears relatively quickly during sleep (4-hour time constant). After 8 hours of sleep, pressure is near zero. After 16 hours awake (a normal waking day), pressure has risen to a moderate level. After 24 hours of continuous wakefulness, pressure is severe.

Process S is the largest single component of the fatigue score. A worker who gets shortened sleep carries residual sleep pressure into their next shift — this is the primary mechanism through which short rest periods create compounding fatigue.

3.2 Process C — Circadian Rhythm

The human body runs on an internal approximately 24-hour clock governed by the suprachiasmatic nucleus. This clock governs alertness independently of sleep history. Process C models this rhythm as a sinusoidal oscillator:

```
C(t) = A × sin(2π(t - ϕ) / 24)
```

Parameters:

```
A = 0.5 (amplitude)
ϕ = 16.8 hours (phase offset — sets peak alertness at ~16:00)
```

- **Circadian trough:** Lowest alertness occurs at approximately 03:00–05:00 hours. Night shifts spanning this window receive a circadian penalty of up to **+16 points** on the fatigue score.
- **Circadian peak:** Highest alertness occurs at approximately 16:00 (mid-to-late afternoon). Day and evening shifts receive a circadian benefit of up to –**9 points**.

Even a fully rested worker experiences a significant dip in alertness during the circadian trough. Circadian rhythms do not adapt quickly to schedule changes — shifting from days to nights takes multiple days of consistent exposure, and many rotating schedules never hold a worker on one pattern

long enough for adaptation to occur.

3.3 Process W — Sleep Inertia (Theoretical)

The published Three-Process Model includes a third component: sleep inertia. When a person wakes from sleep, there is a short but measurable period — typically 20 to 60 minutes — during which performance is temporarily impaired. This is modelled as an exponential decay from an initial impairment value at the moment of waking:

```
W(t) = W■ × exp(-t / τ_i)
```

Published Parameters:

```
W■ = 0.3 (initial inertia magnitude)
τ_i = 0.5 hours (decay time constant)
```

Implementation note: In the Rotation Analytics implementation, Process W is set to zero (W■ = 0). The rationale for this decision is documented in Section 10 (Calibration Notes). The theoretical model is included here for completeness.

4. The Fatigue Score

The active processes combine into a single Fatigue Score on a 0–100 scale:

$$\text{Fatigue Score} = (S \times 100) + C + (D \times 2)$$

Where:

- **S × 100** converts the 0–1 sleep pressure value to a 0–100 base score
- **C** is the Circadian Modifier, ranging from approximately –9 (afternoon, favourable) to +16 (early morning, unfavourable)
- **D × 2** adds 2 points per hour of accumulated sleep debt

The composite score is clamped to the 0–100 range. A score of 0 represents a fully alert worker. A score of 100 represents extreme fatigue. The score represents the estimated fatigue state of a representative worker at any given point in the rotation.

5. Sleep Prediction Model

The model does not assume workers sleep a fixed number of hours between shifts. Instead, it predicts when sleep occurs based on the worker's physiological state and schedule constraints.

5.1 Sleep Onset

The worker falls asleep when sleep pressure is high enough and the body's circadian clock is in a sleep-favourable phase. During multi-day breaks, the worker sleeps each night naturally — the model recognises that once sleep pressure rebuilds to a moderate level during nighttime hours (22:00–06:00), the worker will fall asleep again, producing a normal daily sleep-wake cycle across rest periods.

5.2 Sleep Termination

The worker wakes when sleep pressure has cleared sufficiently, or when the circadian clock enters a strong alerting phase (natural morning waking), or when the next shift is approaching. No fixed sleep window is assumed — sleep timing adapts naturally to the schedule.

Night shift workers are predicted to sleep during the day because their sleep pressure is high after the night shift. Predicted sleep duration under normal conditions falls in the 6–8 hour range. When off-time is shortened, predicted sleep is truncated accordingly.

6. Sleep Debt

When predicted sleep falls below 6 hours and the gap between shifts is short (under 16 hours), the shortfall accumulates as sleep debt. Each hour of debt adds **2 points** to the Fatigue Score.

Sleep debt resets to zero when the worker achieves a full sleep period of **7 hours or more**.

This design is deliberate. Sleep debt is intended to capture the compounding effect of consecutive short-rest shifts — not to penalise workers who are well-rested during long recovery periods. A worker with several days off who sleeps less simply because they are not tired does not accumulate debt. The 16-hour gap threshold ensures that only genuinely compressed rest periods contribute to debt accumulation.

7. Simulation Methodology

The model runs a continuous simulation at 15-minute resolution across every day in the rotation — both worked days and days off.

Time Grid Specification:

```
Resolution: Δt = 0.25 hours (15 minutes)
Example: 14-day rotation = 336 hours = 1,344 time steps
```

Variable	Description
`Time`	Timestamp at 15-minute resolution
`Working`	1 if within shift, 0 otherwise
`SleepOpportunity`	1 if off-shift and in sleep-favourable window
`Sleeping`	1 if predicted sleep is occurring

Variable	Description
`SleepPressure`	Current Process S value (0–1)
`Circadian`	Current Process C value
`FatigueScore`	Composite fatigue metric (0–100)

Table 1: State variables tracked at each 15-minute time step.

• Recovery during days off is visible in the data — you can see sleep pressure drop and fatigue scores decrease across rest periods, confirming the schedule provides adequate recovery.

• The worker's physiological state entering their next shift block is accurately carried forward from the preceding days off, rather than being artificially reset.

• On days off, the fatigue score represents the worker's resting physiological state at the end of that day — effectively, how fatigued they would be if called in at that point.

8. Engine Architecture

The fatigue engine follows a sequential processing pipeline:

1. **Schedule Extraction** — Read shift data from the submitted rotation file
2. **Off-Time Calculation** — Derive off-duty intervals between consecutive shifts
3. **Timeline Generation** — Build the 15-minute resolution simulation grid
4. **Sleep Prediction** — Apply sleep opportunity logic and predict sleep periods
5. **Process S Calculation** — Compute homeostatic sleep pressure at each time step
6. **Process C Calculation** — Compute circadian rhythm value at each time step
7. **Fatigue Scoring** — Combine S and C into the composite Fatigue Score
8. **Sleep Debt Modifier** — Track cumulative debt and apply upward modifier
9. **Risk Classification** — Apply calibrated risk thresholds and generate per-shift flags
10. **Output Generation** — Produce fatigue curves, risk flags, and per-shift scores

All calculations are performed in memory. Expected computation time is under one second for a standard rotation.

9. Risk Classification Framework

Fatigue Scores are mapped to four operational risk categories. These thresholds have been calibrated against real-world shift schedules to produce meaningful, actionable classifications (see Section 10 for calibration rationale).

Score Range	Classification	Description
0 – 44	**Low**	Worker is operating within normal physiological parameters for a standard shift. No intervention
45 – 59	**Moderate**	Elevated fatigue. Typical of evening shifts and consecutive day shifts with mildly shortened slee
60 – 74	**High**	Significant fatigue. Typical of night shifts, where the circadian trough compounds sleep pressur
75 – 100	**Critical**	Severe fatigue. Occurs with extended wakefulness, deep circadian trough, and compounding s

Table 2: Risk classification framework with calibrated thresholds.

In addition to score-based classification, the engine flags:

- Fatigue Score exceeding High or Critical threshold during a shift
- Night shift exposure during circadian trough (03:00–05:00)
- Predicted sleep opportunity less than 6 hours between shifts
- Sleep debt accumulation across consecutive shifts

10. Calibration Notes

The Three-Process Model as published provides theoretical parameter values derived from laboratory studies. Applying these parameters directly to real-world rotation schedules produced classifications that did not align with observed operational reality. Rotation Analytics calibrated two aspects of the model to produce meaningful, actionable results.

10.1 Sleep Inertia (Process W) — Zeroed

The published model includes a sleep inertia component (Process W) that penalises the worker for 20–60 minutes after waking. This process is relevant in on-call and emergency response environments where personnel are woken mid-sleep and must perform immediately.

Rotation Analytics analyses scheduled rotations — not on-call or interrupt-driven schedules. In a scheduled rotation, workers wake before their shift and have time to clear sleep inertia before reporting. Including Process W would penalise the start of every shift for a physiological state that has resolved by the time the worker is on duty.

Accordingly, W■ is set to 0 in the current implementation. If on-call or emergency response schedules are analysed in the future, Process W can be activated with published parameters.

10.2 Risk Classification Thresholds — Calibrated

The published model suggests risk thresholds at lower Fatigue Score values (approximately 0–29 Low, 30–49 Moderate, 50–69 High, 70–100 Critical). When applied to real-world healthcare rotation schedules, these thresholds classified the majority of standard shifts as Moderate or High, including schedules that are well-established and operationally sound.

This overshoot occurs because laboratory-derived parameters do not fully account for the adaptation that real workers achieve through consistent scheduling patterns. To produce classifications that distinguish genuinely concerning schedules from standard shift work, the thresholds were recalibrated against actual rotation data:

- Schedules known to be operationally acceptable were verified to classify as Low
- Evening shifts and mildly compressed schedules were verified to classify as Moderate
- Night shift blocks with circadian trough exposure were verified to classify as High
- Only genuinely extreme schedules — extended wakefulness with compounding debt — reach Critical

The recalibrated thresholds (0–44 / 45–59 / 60–74 / 75–100) produce risk classifications that align with operational experience and clinical expectation. This calibration is what makes the analysis actionable rather than merely theoretical.

11. Model Assumptions and Limitations

11.1 What the Engine Can Do

- Predict fatigue levels for a representative worker based on a given schedule
- Identify shifts and time windows that carry elevated physiological risk
- Compare the fatigue profiles of different scheduling options objectively
- Detect cumulative fatigue accumulation across a rotation that point-in-time rules miss

11.2 What the Engine Cannot Do

- Account for individual differences in sleep need, chronotype, or health conditions
- Measure actual fatigue — only predict it based on schedule inputs
- Replace clinical or occupational health assessment for a specific individual
- Guarantee that a low-risk schedule will produce no fatigue-related incidents

11.3 Assumptions

- The model represents a **typical working-age adult**. Individual variation requires individual data (actigraphy, sleep diaries) that is not available in a scheduling context.
- The model assumes the worker follows the **scheduled shift pattern as provided**. It does not account for overtime, shift swaps, or absences.
- Results should be interpreted as **relative risk indicators**, not absolute performance predictions.
- The analysis is informational and **does not constitute medical, safety, or legal advice**.

12. Deliverable Format

Fatigue analysis results are delivered alongside compliance findings. The fatigue deliverable includes:

- **Per-line fatigue scores:** average, minimum, and maximum scores for each rotation line.
- **Risk classification** for each line based on average and maximum scores.
- **Rotation-wide trend visualisations** (sparklines) showing the complete fatigue and recovery cycle across every day of the rotation — both worked days and days off.
- All sparklines use the same **0–100 vertical scale** and are directly comparable across rows.
- **Risk flags** for night shift circadian trough exposure, shortened sleep opportunity, and sleep debt accumulation.

13. Mathematical Appendix

Complete mathematical specification of the model as implemented.

A. Process S — Homeostatic Sleep Pressure

During wakefulness:

```
S(t) = S(t-1) + (1 - S(t-1)) × (Δt / τ_w)
```

During sleep:

```
S(t) = S(t-1) - S(t-1) × (Δt / τ_s)
```

```
τ_w = 18 hours τ_s = 4 hours Δt = 0.25 hours
```

B. Process C — Circadian Rhythm

```
C(t) = A × sin(2π(t - ϕ) / 24)
```

```
A = 0.5 ϕ = 16.8 hours
```

C. The Fatigue Score

```
FatigueScore(t) = (S(t) × 100) + C(t) + (SleepDebt × 2)
```

```
FatigueScore(t) = max(0, min(100, FatigueScore(t)))
```

D. Sleep Debt

```
IF PredictedSleep < 6h AND ShiftGap < 16h:
SleepDebt += (6 - PredictedSleep)
IF PredictedSleep >= 7h:
SleepDebt = 0
```

E. Sleep Prediction

```
Sleep Onset: Off-shift AND SleepPressure high AND
CircadianPhase in sleep-favourable window
Sleep End: SleepPressure cleared sufficiently OR
Circadian alerting phase OR
Next shift approaching
```

F. Process W — Sleep Inertia (Zeroed)

```
W(t) = W■ × exp(-t / τ_i)
W■ = 0 (zeroed in current implementation)
Published: W■ = 0.3, τ_i = 0.5 hours
```

14. References

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